

S06 catalog publication and recovery

Internal Rust storage feature. No weather route, provider request, production resource or deployment is enabled by this task. The example uses a fresh local test S3 peer with obvious dummy credentials; it does not verify AWS SigV4, IAM, Versioning, lifecycle, response parity or AWS latency.

Run the actual HTTP scenario

From server2/, with the configured Rust toolchain and Python3:

```
cargo build --release --example catalogs_smoke
python3 tests/storage/catalog_peer.py --smoke target/release/examples/catalogs_smoke
```

The public peer starts an ephemeral IPv4 loopback port, runs the release example, and closes its server/child resources. Both the reusable fault peer and release scenario are maintained here. The captured usage image shows actual stdout from this command, not a simulated output. Detailed Rust wire assertions run through `cargo test --workspace`; the example is a separate functional execution.

Cold read and concurrent writes

Prepare a complete `GroupDeclaration` before publishing any raw body. Members contain canonical S05 envelopes with `fetch_group:None`, ordered pages and exact sorted catalog ownership. Hash the declaration, ordered members and partitions; attach its resulting immutable reference before each raw PUT. Store the descriptor first, then all raw bodies, then union each catalog using opaque ETag/If-Match. Only all-sibling and exact raw verification returns a checked complete acquisition. A raw PUT acknowledgment is body durability only.

Catalog keys are `index/v2/{source}/{kind}/{key_sha256}/{partition}.json`. Descriptors use `index/v2/groups/{group_sha256}.json`. JSON stores identity/envelope metadata only. Every revision remains; arbitrary normalized values are rejected by typed schemas. Group SHA has no self-reference because declared envelopes exclude group refs. Pages 1..N must be complete, unique and ordered. Previously ungrouped/differently grouped raw identities cannot be retroactively affiliated.

Whole-history Lookup pins each involved catalog once. `CompleteSet` exposes complete acquisitions and every dependency identity, including absent siblings through optional generation/ETag tokens. `A/new` plus `B/healthy-old` excludes the new group. Old complete acquisitions may remain usable with `excluded_groups()>0`; consumers must schedule re-lookup on exclusions, including missing descriptor/body cases, without assuming unchanged target ETag proves complete coverage. S07 owns this cache refresh/invalidation policy. Earliest/latest, absent-only top-level field merge and whole-list replacement are derived in memory from ordered complete acquisitions; present null/zero/false fields are never overwritten by absent merge.

Bounded history reads and publication scope

Archival catalogs retain every revision. Normal read-budget exhaustion is Capacity or Timeout, not corruption and not a provider miss. Full Lookup is the only `FullHistory CompleteSet`; only that coverage permits fold. Publication verifies only the newly declared group and returns `PublishedGroup` coverage; attempting to fold that scoped result as history is rejected. Parse its explicit records directly when the desired operation is that single acquisition.

Use `lookup_page(id, order, cursor, maximum_acquisitions, deadline)` for bounded whole-acquisition pages ordered exactly like `FullHistory`: first normalize each unique group using its verified descriptor's ordered first member, even when that member belongs only to another partition, then sort by fetched time/full raw hash/canonical identity and apply the requested direction. Raw-member extremes do not define the acquisition order. Pages have `Paged` coverage and no fold method. Ordered provider pages remain together. The opaque private cursor pins target generation/ETag/version, prior sibling dependencies and a digest of the normalized ordering cohort (representative identity plus full group reference). If a missing descriptor appears without changing the catalog ETag, the cohort change also forces `Incomplete/restart`; cross-ID/order use is invalid and changed tokens yield `Incomplete/restart`. Consumers must not merge one page as all history. Collect every page and revalidate dependencies before a cross-page historical result; growing dependency checks may still exceed the deadline or cursor metadata cap, requiring safe fallback. Metadata normalization performs upfront bounded descriptor GETs across the target catalog; only selected groups load raw bodies/sibling catalogs. This increases page lookup cost and may reach Capacity/Timeout; a one-item page does not promise three GETs independent of retained history. Exact targeted lookup below avoids that normalization.

`lookup_acquisition(id, exact_member_identity, deadline)` is the independent escape hatch: verify target membership and that member's whole immutable group/siblings. It returns Targeted coverage with no continuation, which does not mean exhausted history. Old/new acquisitions remain accessible within their own group/catalog bounds without reading unrelated historical bodies. Sequential reads and S07/S09 gates remain.

Preflight rejects an oversized new group with Capacity before any PUT. It checks raw total, descriptor, projected catalog union and conservative final metadata: descriptor bytes + partition count × control-document limit must fit the retained control cap. Format allows 16 partitions, but default admission permits at most 7 full-size catalogs plus descriptor; smaller store caps may admit fewer. No archive pruning is used. Preflight buffers are dropped before CAS, and completed CAS-phase buffers before fresh new-group pinning. Control accounting is bounded per retained phase, not cumulative wire bytes across all phases; CAS work and the outer deadline still bound retries. Within the CAS phase, conflict re-reads are charged cumulatively even when earlier buffers were released, so sustained contention can safely return Ambiguous. Concurrent growth or uncertain sent writes may also remain Ambiguous.

Repair and coverage

Call `repair` with an explicit `RepairScope` containing the target catalog and all candidate canonical raw periods/slots for that provider resolution. It scans even healthy old catalogs, follows every LIST page, verifies raw key/hash/envelope and full group descriptor/membership for every discovered envelope, then CAS repairs all required catalogs and re-pins siblings. Validated descriptors and expected envelopes are cached under the control/deadline limits, but every raw row must match exact membership, ownership and its complete immutable group reference before deduplicating full-group reload or CAS. Descriptor caching does not itself authorize indexing: at least one exact member owned by the target catalog must be discovered before verifying and repairing the complete group. Invalid-only or foreign-only claims receive no group catalog PUT. A stray row claiming an already-seen valid group keeps the scope Incomplete. It does not prevent independently verified complete orphan groups from being indexed; only the descriptor-declared members are recovered. A fresh exact targeted lookup can read those safe groups, while the mixed scope never becomes Complete, Indexed or CompleteEmpty. Indexing an acquisition does not prove complete scope coverage. A partial LIST, malformed XML/token, missing/corrupt member, wrong raw hash, deadline or cap never becomes a complete empty response.

`LookupOutcome::Incomplete` is not provider admission. `RepairOutcome::CompleteEmpty` proves absence only in the caller-supplied finite scope after an exhausted clean scan; it does not prove global archive absence. `Complete` contains verified data; `Indexed {scope, dependencies}` records an exhausted, verified nonempty scan and successful identity-index repair when full-history materialization exceeds capacity. It is neither a serving `CompleteSet` nor `CompleteEmpty` and cannot authorize providers; read checked pages or targeted acquisitions afterward. A materialization deadline can still yield Timeout/Ambiguous rather than a positive Indexed signal. Incomplete or errors leave valid memory/legacy error fallback. Discovery retains envelopes only; verified body buffers are released before reloading each complete group, without double-counting discarded bodies as retained output. No provider calls or budget reservations occur in this utility.

The caller must map every supplied period to the target partition deterministically. Foreign-owned grouped records are Incomplete; ungrouped singletons rely on this caller ownership assertion. This utility cannot infer a geocode/calendar layout. Default scope has up to 16 exact canonical periods/slots. Later geocode integration must supply its approved month-based/projection identities and prove 30-day validity plus multi-month restore (including January 31 to March 1); S06 has not executed that geocode fixture. Calendar/coverage policy and provider parsers remain later work. A descriptor with no discoverable raw body cannot be found by raw-prefix scan alone; known indexed missing bodies remain incomplete. No claim of global orphan absence is made from direct catalogs or finite scans.

Failures and limits

Completed conditional PUT 200 is the only write acknowledgment. HTTP 412 causes re-read/union; HTTP 409 and unknown writes reconcile within a bounded attempt count. A sent unknown PUT remains uncertain through a later read 404/retry rejection: return Ambiguous unless exact stored identity/union proves success. A first direct 403 stays `Status(403)`; malformed identities/corruption fail closed. Cancellation or an operation deadline after publication begins cannot return a complete group. No hidden SDK retry or third provider acquisition is used.

Defaults: JSON/XML 1 MiB, catalog 2,048 entries, group 128 members, 16 partitions, 8 LIST pages per prefix, 2,048 raw candidates, 1,000 keys/page, 4 CAS attempts, 8 MiB serialized control budget per retained phase and 32 MiB decoded serving output. Repair discovery discards raw bodies, and each group reload/output is bounded separately. Legal chunked or close-delimited control responses may omit Content-Length; streaming caps remain enforced and any declared length must match. Empty truncated LIST pages follow valid opaque tokens under existing page and repeated-token bounds. The caller's

absolute monotonic deadline is capped by the store's 3 s default; all credential waits, siblings, raw reads, CPU work and CAS/LIST count toward it. Hard setter ceilings fail safely. Bounded I/O operation admission and CPU workers remain held for started parsing work after cancellation. Within one lookup/repair, descriptor and raw-member GETs currently run sequentially. The I/O limit of 16 bounds concurrent operations; it does not produce 16 parallel GETs inside one lookup. The architecture cold-latency model assumes prospective bounded fan-out, not this implementation. S07 integration and S09 feasibility/cutover measurements must resolve this limit before a cold weather route is switched. No AWS latency or complete-day timing is established by the small local smoke. Limits are per store/operation, not hard process RSS; caller input/results and concurrent stores consume memory.

Real AWS role 404 versus 403, signature/header behavior, Versioning/no-delete policy and performance remain staging gates. No delete/lifecycle/pack work is implemented. Catalog current revisions are retained; version-LIST is unnecessary on this path. Oversized catalogs fail rather than silently pruning revisions. Route cutover and cold-cell parity/fallback gates remain unchanged.

Verification artifacts

The maintained peer's concurrency fault holds the first two conditional catalog PUTs before the atomic condition check and commit; preflight GETs remain normal. The wire regression checks actual per-key 412 response counts, at least three catalog PUTs and both complete acquisition groups over ten fresh peers. A missing second participant returns an explicit test 503 before the inherited socket lifetime ends. This fault is test infrastructure, not a production retry or admission policy. Response counters use a separate lock because conditional errors can be emitted while the peer holds its object lock. The workspace still runs with default test parallelism; it is not serialized to hide races.

The deadline-resume fixture first proves exact descriptor/raw bytes and envelopes, catalog A and absent B after a direct B PUT403. Its 70ms resume expires in preflight with Timeout and zero additional PUTs; clearing the fault restores the complete whole group. It does not assume a short deadline already committed the raw inputs. Sent-write Ambiguous and cancellation regressions remain separate checks. History fixtures keep every published acquisition and cursor page. Explicit byte/pin reader limits require exact Capacity after one raw GET before a default history read; the default read may hit only documented Capacity or Timeout. Successful metadata-page ordering/count coverage clears the synthetic delay, which is restored for the deadline check. These test settings change no default production limits or stored revisions.

The editable manual and rendered PDF share this content. The selected usage capture and hash manifest live under docs/evidence/tasks/server2-catalogs/. The release scenario checks cold complete pages, A/new+B/old exclusion, healthy multi-page repair, committed response loss, delayed first commit with retry403, first direct403, bounded ordered pages/targeted scope/fold guard, chunked LIST, empty truncated continuation, same-fetch/hash-tied group chronology across cursor pages, and rejection of undeclared raw membership after seeing a valid group. See the [architecture publication contract](#) and [publication sequence](#).

Actual local CLI usage

```
S06 - actual release HTTP smoke | loopback only

$ python3 tests/storage/catalog_peer.py --smoke target/release/examples/catalogs_smoke

publish: complete group, 2 ordered raw pages, 2 catalog partitions
cold: 1 checked acquisition, 2 pinned dependencies, providers disabled
partial: A/new + B/healthy-old excluded; preceding complete group retained
repair: healthy catalog + paginated LIST restored every revision
response loss: committed CAS reconciled from exact union
late commit: lost first PUT + read404 + retry403 remains Ambiguous; exact catalog exists
direct rejection: first PUT403 retained, no group success
bounded read: whole 2-page acquisition, opaque continuation and exact targeted scope; scoped publication
fold blocked
streamed repair: chunked LIST and empty truncated continuation exhaust only the explicit empty scope
acquisition order: tied fetch/hash groups match FullHistory across latest-first cursor pages
repair membership: undeclared raw claiming an already-seen valid group stays Incomplete
mixed repair: scope stays Incomplete while verified orphan is indexed and cold exact lookup succeeds
PASS local real HTTP smoke (222.67 ms); not AWS latency/auth or API parity
```